

EXPERIMENT – 06

Aim: Design and train a LSTM model for real world application of your choice.

Theory:

Overview

Stock price prediction is a challenging task due to the inherent randomness and complexity of financial markets. Traditional machine learning models struggle to capture the temporal dependencies in stock price movements. Long Short-Term Memory (LSTM) networks, a type of recurrent neural network (RNN), are well-suited for sequential data processing and can effectively capture trends and patterns in time-series data.

LSTM networks are a special kind of RNN capable of learning long-term dependencies in data. Unlike traditional RNNs, which suffer from the vanishing gradient problem, LSTMs use memory cells and gating mechanisms (input gate, forget gate, and output gate) to retain important information over long sequences.

Key advantages of LSTM for stock price prediction:

- Ability to learn sequential dependencies in data.
- Capturing long-term trends and patterns in stock prices.
- Handling noisy and non-stationary financial data.

Dataset and Preprocessing

For this project, we use stock price data obtained from Yahoo Finance. The dataset includes various attributes such as Open, High, Low, Close, Volume, and Adjusted Close prices. The **closing price** is chosen as the target variable for prediction.

Steps in Data Preprocessing:

1. **Data Collection:**
 - Using the `yfinance` library, we fetch historical stock price data for a given period.
2. **Feature Selection:**
 - We extract the closing price as it represents the final price of the stock for each trading day.
3. **Normalization:**
 - Stock prices are scaled using **MinMaxScaler** to normalize them between 0 and 1. This improves the model's learning efficiency.
4. **Creating Sequences:**

- LSTMs require sequential data as input. A sliding window approach is used to create sequences of past stock prices for training the model.
5. **Splitting the Dataset:**
- The dataset is divided into **training** and **testing** sets, typically using an 80:20 ratio.

LSTM Model Architecture

The LSTM model is implemented using TensorFlow/Keras with the following architecture:

1. **Input Layer:**
 - Takes in sequences of stock prices of length 60 days.
2. **LSTM Layers:**
 - Two LSTM layers are used, with 50 neurons each. The first LSTM layer returns sequences for the second LSTM layer.
3. **Dense Layers:**
 - A fully connected Dense layer with 25 neurons and ReLU activation.
 - The output Dense layer predicts the next day's stock price.
4. **Compilation:**
 - The model is compiled with Adam optimizer and Mean Squared Error (MSE) loss function.
5. **Training:**
 - The model is trained using the training dataset with batch size 32 for 5 epochs.

Making Predictions

After training, the model is used to make predictions on the test data. The predicted values are inverse transformed to their original scale using the `MinMaxScaler`. The results are plotted to compare actual vs predicted prices.

Results and Observations

- The model successfully captures overall trends but may struggle with sudden spikes or dips.
- Accuracy depends on the quality of input data and hyperparameter tuning.
- More complex models with additional features like trading volume, moving averages, and external economic indicators may improve predictions.

Conclusion

LSTM networks provide a powerful approach to stock price prediction by capturing sequential dependencies in financial data. However, stock prices are influenced by many external factors such as economic news, global events, and market sentiment. Combining LSTM with sentiment analysis or other deep learning techniques can further enhance prediction accuracy.

Future Improvements:

- Using additional financial indicators such as RSI, MACD, and Bollinger Bands.
- Implementing Attention Mechanisms for better feature weighting.
- Integrating sentiment analysis from financial news and social media.
- Increasing model depth and fine-tuning hyperparameters for better performance.

By leveraging deep learning techniques, we can build robust models that assist traders and investors in making informed decisions.